FIVE EQUIVALENT FORMULATIONS OF QUASI-COMPACTNESS FOR ALGEBRAIC STACKS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1387-1402 of sources/stacks-stacks-properties/pinned-original-source.tex; proof span is lines 1404-1422, the complete author proof (17 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 21 of queue_stacks-properties.json (pb=967, stmt 1387-1402, proof 1404-1422).

Editorial note. stacks-properties wave-1: fresh adjudication accept (2026-10-14); pool scanned while CONTROL.md is down (0 bytes) - §2 row deferred per Chief, claim reported to Chief 08:0x.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma lemma-characterize-surjective: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 1051–1058. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01I5: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 1422–1422. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. The following are equivalent:*

(1) *$\mathcal{X}$ is quasi-compact,*
(2) *there exists a surjective smooth morphism $U \to \mathcal{X}$ with U an affine scheme,*
(3) *there exists a surjective smooth morphism $U \to \mathcal{X}$ with U a quasi-compact scheme,*
(4) *there exists a surjective smooth morphism $U \to \mathcal{X}$ with U a quasi-compact algebraic space, and*
(5) *there exists a surjective morphism $\mathcal{U} \to \mathcal{X}$ of algebraic stacks such that $\mathcal{U}$ is quasi-compact.*

Date: stacks-project:stacks-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. We will use Lemma lemma-characterize-surjective. Suppose $\mathcal{U}$ and $\mathcal{U} \to \mathcal{X}$ are as in (5). Then since $|\mathcal{U}| \to |\mathcal{X}|$ is surjective and continuous we conclude that $|\mathcal{X}|$ is quasi-compact. Thus (5) implies (1). The implications (2) $\Rightarrow$ (3) $\Rightarrow$ (4) $\Rightarrow$ (5) are immediate. Assume (1), i.e., $\mathcal{X}$ is quasi-compact, i.e., that $|\mathcal{X}|$ is quasi-compact. Choose a scheme U and a surjective smooth morphism $U \to \mathcal{X}$. Then since $|U| \to |\mathcal{X}|$ is open we see that there exists a quasi-compact open $U' \subset U$ such that $|U'| \to |X|$ is surjective (and still smooth). Choose a finite affine open covering $U' = U_1 \cup \ldots \cup U_n$. Then $U_1 \amalg \ldots \amalg U_n \to \mathcal{X}$ is a surjective smooth morphism whose source is an affine scheme (Schemes, Lemma 01I5). Hence (2) holds. $\square$